Local Similarity-Based Spatial–Spectral Fusion Hyperspectral Image Classification With Deep CNN and Gabor Filtering

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***Abstract*— Currently, the different deep neural network (DNN) learning approaches have done much for the classification of hyperspectral images (HSIs), especially most of them use the convolutional neural network (CNN). HSI data have the characteristics of multidimensionality, correlation, nonlinearity, and a large amount of data. Therefore, it is particularly important to extract deeper features in HSIs by reducing dimensionalities which help improve the classification in both spectral and spatial domains. In this article, we present a spatial–spectral HSI classification algorithm, local similarity projection Gabor filtering (LSPGF), which uses local similarity projection (LSP)-based reduced dimensional CNN with a 2-D Gabor filtering algorithm. First, use the local similarity analysis to reduce the dimensionality of the hyperspectral data, and then we use the 2-D Gabor filter to filter the reduced hyperspectral data to generate spatial tunnel information. Second, use the CNN to extract features from the original hyperspectral data to generate spectral tunnel information. Third, the spatial tunnel information and the spectral tunnel information are fused to form the spatial–spectral feature information, which is input into the deep CNN to extract more effective features; and finally, a dual optimization classifier is used to classify the final extracted features. This article compares the performance of the proposed method with other algorithms in three public HSI databases and shows that the overall accuracy of the classification of LSPGF outperforms all datasets.**

***Index Terms*— Convolutional neural network (CNN), deep convolutional neural network (DCNN), Gabor features, hyperspectral image (HSI) classification.**

I. Introduction

NOWADAYS, hyperspectral image (HSI) analysis has been widely used in different fields [1], such as agriculture and land change monitoring [2], [3], geographical science for atmosphere, and mineral development [4]. The hyperspectral sensor captures hundreds of 2-D ground space images on continuous narrow bands to generate 3-D HSI containing spectral and spatial information [5]. Hyperspectral sensors provide a continuous spectrum curve that many traditional spectral sensors cannot provide. Its wavelength ranges from visible light to infrared light [6] and contains rich spectral information. Although the rich spectral information and spatial information of HSI are conducive to the classification of ground targets, they also face certain difficulties in practical applications.

1) First, HSIs have typical characteristics of high dimensionality and high information redundancy. Compared with ordinary images, they contain a larger amount of information, which results in a large number of calculations for processing a certain pixel in the classification and require dimensionality reduction (DR) preprocessing on HSIs.
2) Second, the number of label samples used for classification in HSI data may be small, because the acquisition of label samples consumes a lot of resources. But in the process of HSI classification, only a small number of label samples can be used in the training stage of classification [11]. Therefore, it is very important to study to achieve accurate image classification with a small number of samples with good selection of features. And transfer learning has been proven to have outstanding advantages in completing the target tasks in new fields with insufficient sample size. Therefore, studying the classification method of HSI based on transfer learning is of great significance to solve the problem of low classification accuracy under high-dimensional small samples.

Classification of HSIs can be divided in two categories spatial–spectral and pixel-wise approach [42]. In pixel wise use of spectral signatures help in classification while in spatial–spectral exploits the spatial and spectral information. In HSI classification, spectral based classification is commonly used to obtain better results and they can be further classified into two categories traditional approach and machine learning approach. In traditional approaches, principal component analysis (PCA) has become a commonly used method to reduce high-dimensional data to low-dimensional data [7]. There are other approaches also which help in dimension

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reduction such spectral angle mapper (SAM) [43], minimum noise fraction (MNF) [44] are often used for spectral features extraction. After feature extraction these features were sent to classifiers like k-nearest neighbors (KNN) [45], support vector machine (SVM) [36], random forests (RFs), and so on. Spectral-based classification using machine learning are gradually raised and typical methods stacked autoencoder (SAE) [38], 1-D convolutional neural network (CNN) [11], and multi feature learning (MFL) [39]. However, only extracting spatial features or spectral features cannot fully integrate spatial and spectral information [8], and the fusion of spatial–spectral information is necessary to improve the classification accuracy of HSI. Results of spectral classification can be further improved when combined with spatial classification [40]. Veganzones *et al.* [46] developed a region-based merging algorithm using spectral–spatial regions that are clustered to the segment image. Liu *et al.* [42] used 3-D Gabor filtering for spatial–spectral feature extraction HSI classification and decomposed in low- and high-pass filters to get more accurate classification results. The method of using PCA to extract the first three principal components not only achieves data DR but also integrates spatial spectrum information to improve classification accuracy [9]. Also, the information entropy theory quantifies the information of each band and sorts it according to the information gain [10]. Therefore, researches on the DR of HSI data and the realization of spatial spectral information fusion based on the information measurement theory are important for improving the accuracy of HSI classification and have a very important meaning.

The categories of HSI are complex and diverse, and the classification of pure spectral information or pure spatial information is difficult to apply to all types of hyperspectral data. At present, it is generally believed that the simultaneous use of spatial and spectral information is of great help to improve classification accuracy. Extracting the spectral and spatial information contained in the original data and using different types of features for spectral–spatial classification can effectively improve the classification accuracy. Therefore, making full use of spatial information and spectral information for pixel classification has become a research hotspot in HSI classification. In recent years, researchers have proposed many space–spectral combined HSI methods [23], [24].

Also, as an advanced machine learning technology, deep learning (DL) uses deep neural networks to learn hierarchical features from raw input data from low to high levels, and it has been widely used in HSI classification [12]. Traditional feature extraction methods are mainly manual extraction methods, which may not meet the needs of embedding detailed features in actual data. It is very challenging to achieve the best classification accuracy and robustness balance in many hyperspectral data [13]. The model based on deep convolutional neural networks (DCNNs) can detect the local features of the input hyperspectral data, to obtain high accuracy and stable classification results. From a spatial perspective, the pixels that are adjacent in space are most likely to belong to the same category, i.e., they have spatial consistency. To make full use of spatial features, various methods of extracting spatial features have been proposed, such as gray-level co-occurrence matrix [20], wavelet transform feature extraction [21], extended morphological contour [22], and so on. These methods need to manually set parameters in advance when extracting spatial features. The extracted spatial features can only be targeted at specific target objects, lacking flexibility, and low-level features, such as shape, texture, and edges, vary greatly, making these methods impossible to set experience parameters to describe all types of objects.

For data DR, scholars have proposed many classic algorithms using different techniques of machine learning. These methods usually high-dimensional spectral vectors to low-dimensional feature spaces based on certain criteria and then classify them in low-dimensional feature spaces [14]. Classical methods include linear discriminant analysis (LDA) [15], Kernel principal component analysis (KPCA) [16], Kernel Fisher discriminant analysis [17], generalized discriminant analysis [18], and SVMs [19]. Although the abovementioned algorithm has achieved good results, it still has shortcomings. LDA assumes that the class conditional distribution is a Gaussian distribution, while hyperspectral data is often a non-Gaussian distribution, and may even be multimodel, and the local characteristics of adjacent pixels in the input space cannot be well protected in the process of maximizing the interclass distribution matrix which leads to loss of information. Local similarity preserving based project can effectively solve many dimension reduction issues because of its predefined calculation of similarity among neighbor pixels and adaptive selection of sample class samples data [13].

DL has made more and more achievements in the field of image classification. Compared with the traditional manual setting of spatial features, it can automatically extract spatial features hierarchically and shows higher effectiveness and stronger robustness in image classification. In the field of remote sensing classification, DL shows great potential [25]–[27]. The above methods use a layered structure to extract deep spectral features from the original spectral information to complete the classification of hyperspectral remote sensing data while ignoring the spatial feature information contained in the original data. Gabor wavelet has good spatial locality, frequency selectivity, and direction selectivity. It can extract features that reflect the frequency and direction information of the image, especially the local area. It has achieved better results in face recognition and texture classification [28]. The spatial consistency of hyperspectral remote sensing images makes the images of similar features have greater similarity in both spatial frequency and directionality, and these spatial characteristics of different features have greater differences. The 2-D Gabor filtering can extract the spatial characteristics of hyperspectral remote sensing images as well. To make full use of the spatial and spectral information of hyperspectral remote sensing images, this article combines the characteristics of hyperspectral data and DL, and fuses the spectral–spatial combination of HSIs information and extract effective features, and propose local similarity projection Gabor filtering (LSPGF) an HSI classification algorithm based on local similarity projection (LSP), CNN, and Gabor filtering.

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The remainder of this article is organized as follows. Section II describes the proposed algorithm with its related work of implementation. Section III shows the experimental results, dataset details, parameter setting, and different types of evaluation metrics approach. In Section IV, we conclude the results and highlight the contribution of research with some future remarks.

II. Proposed Method

This article on the spectral–spatial characteristics of HSIs based on the local retention DR CNN classification algorithm uses the local similarity DR method to protect the local information of neighboring features and uses the 2-D Gabor filter to extract multiple models of HSI structures; the DCNN framework can extract higher level, more abstract, and more stable features. The algorithm steps in this article are as follows:

1) Preprocess the original HSI and use local retention to measure the spectral dimension of the hyperspectral data.
2) Using the 2-D Gabor filter to extract the spatial features of the HSI.
3) Using the spectral feature extraction tunnel to extract the spectral features of the HSI.
4) Connect the spatial and spectral features and then input them into the fully connected layer.
5) Use the DCNN model to train the processed features.
6) Put the set after DCNN training into the dual-optimized classifier for classification.

The algorithm flow is shown in Fig. 1. Details of the abovementioned steps are as follows.

A. LSP and DR

Each spectral image of an HSI is a mixture of multiple endmember spectral images. For the HSI classification algorithm, it is better to find a low-dimensional subspace to represent a single endmember, which represents the ground features. Classical PCA (component analysis, PCA) and LDA DR methods are difficult to achieve, while the local retention DR method finds a suitable linear map M which makes the adjacent pixels of the original image by having a relatively short projection space distance, thereby effectively retaining the relevant information in the local neighborhood. For HSIs, the statistical structure is more complex, often like non-Gaussian distribution or multimodel distribution. LSP has a local protection constraint, which can effectively protect the local structure of the original HSI. Suppose the training sample of the original data is $x_i = (x_{i,1}, x_{i,2}, \ldots, x_{i,j}, \ldots, x_{i,n})$ and $x_{i,j} \in R^d$, then the class label is $y \in \{1, 2, \ldots, c$, where c represents classes of the sample and n is the training sample total number. Assuming that n_l represents the number of training samples of the class L, then $\sum_{l=1}^{c} n_l = n$. The close relationship between the training sample x_i and the training sample x_j is expressed as

$$A_{i,j} = \exp\left(-\frac{\|x_i - x_j\|^2}{\gamma_i - \gamma_j}\right) \tag{1}$$

among them

$$\gamma_i = \|x_i - x_{i,m}\|$$

where

γ_i, γ_j are the local scale of samples x_i, x_j;

$x_{i,m}$ is the mth adjacent sample of sample x_i.

Based on the original LDA, local preservation and DR obtain a linear map M through LSP to preserve the local information between adjacent pixels [17]. The obtained linear graph M can not only separate the classes well, but also protect the local information within the classes. The local interclass scatter matrix L_{lb} and the intraclass scatter matrix L_{lw} in the locally preserved DR are defined as

$$L_{\text{lb}} = \frac{1}{2}\sum_{i,j=1}^{n} W_{i,j}(x_i - x_j)(x_i - x_j)^T \tag{2}$$

$$L_{\text{lw}} = \frac{1}{2}\sum_{i,j=1}^{n} W_{i,j'}(x_i - x_j)(x_i - x_j)^{\text{T}} \tag{3}$$

where

$W_{i,j}$ is the close value between training samples between classes;

$W_{i,j'}$ is the close value between training samples in the classes.

Among them, the interclass scatter matrix L_{lb} and the intraclass scatter matrix L_{lw} are both $n \times n$ square matrices.

In (1), the close value of adjacent pixels of the same kind is obtained, and the interclass scatter matrix L_{lb} and the intraclass scatter matrix L_{lw} hardly affect the same nonadjacent pixels. By locally spreading the matrix, the maximized Fisher ratio is

$$L_{\text{lb}}^T = \delta L_{\text{lw}}^T \tag{4}$$

where δ is diagonal eigenvalue matrix.

Thus, the transformation matrix is

$$T_{\text{LFDA}} = \arg\max_{T \in R^{D \times r}} \text{tr}\left[\left(L_{\text{LFDA}}^T L_{\text{lb}} T_{\text{LFDA}}\right)^{-1} L_{\text{LFDA}}^T L_{\text{lb}} T_{\text{LFDA}}\right] \tag{5}$$

where tr(·) is matrix trace calculation function.

Local similarity retention and DR have local protection constraints, which can effectively protect the diversified local structure of the original HSI. The constraint is measured by the similarity of local adjacent pixels. Through the transformation matrix T_{LFDA}, the adjacent data of the same type are close to each other, the adjacent data of different types are separated from each other, and the local characteristics of the data are well preserved. Using LSP with DR not only obtains a more compact distribution, but also reduces the Bayesian error, and finally can help achieve the goal of improving the classification accuracy.

B. Spatial–Spectral Feature Extraction

The 2-D Gabor transform function is a sinusoidal function modulated by a Gaussian envelope, and its function expression is [17]

$$g(x, y; \vartheta, \theta, \varphi, \sigma, \gamma) = \exp\left(-\frac{x'^2 + \gamma^2 \gamma'^2}{2\sigma^2}\right)\exp\left(j\left(2\pi\frac{x'}{\delta\vartheta} + \varphi\right)\right) \tag{6}$$

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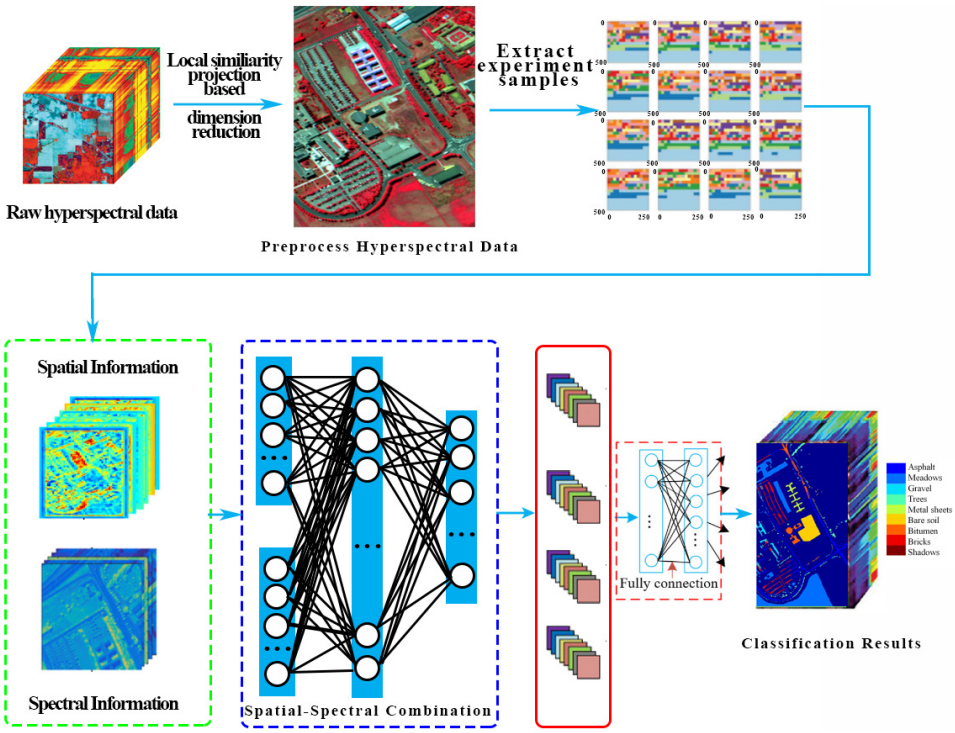


Fig. 1. Proposed model's step-by-step implementation.

where

$$x' = x\cos\theta + y\sin\theta \tag{7}$$

$$y' = y\cos\theta - x\sin\theta \tag{8}$$

where ϑ is the sine factor wavelength;

x, y are the horizontal and vertical coordinate values in the coordinate system;

θ is the angle of the core direction, which specifies the direction of the parallel shift of direction;

φ is the phase offset; the default value $\frac{2}{\pi}$ means that Gabor transform is a complex number with real and imaginary parts;

γ is the space phase ratio that limits the degree of deviation from circularity;

σ is the standard deviation of Gaussian envelope, which is determined by wavelength and spatial frequency bandwidth.

To improve the utilization of spectral data features, a spectral feature extraction tunnel is designed. The spectral feature extraction tunnel focuses on the central pixel Z_{ij} at the P_{ij} position. After convolution and mixing, the output spectral characteristics transform the 3-D spectral feature vector into a 1-D vector [29], which is the depth spectral feature obtained by the algorithm. The spectral tunnel is centered around pixel P_{ij}, and r is the radius as the input data. The original data $Z_{ij} = R^{k\times k}(k = 2R + 1)$ are input into the DCNN tunnel and 2-D convolution, and batch normalization realizes the spectral feature extraction of the area around the center target pixel point [30]. Since the gradient of the unsaturated nonlinear function [rectified linear unit (ReLU)] is unsaturated when $x > 0$, the gradient of the function is always 1. In the backpropagation process, the gradient dispersion problem is alleviated, so that the parameters of the first few layers of the neural network can be updated very quickly; so, the nonlinear function (ReLU) as the activation function not only has a fast convergence speed and a fast training speed, but also a high classification accuracy.

The HSI spatial–spectral feature extraction model designed in this article can fully extract the spatial–spectral features of the HSI and can protect the local information of adjacent features and the multimodel structure of the HSI and improve the classification accuracy and kappa coefficient, especially in the Gaussian noise environment, with excellent classification performance. The spatial–spectral feature extraction model is shown in Fig. 2.

C. Spatial–Spectral Feature Fusion

The spectral features after the tunnel transformation are extracted by using spectral feature classification, and the following equation is used to combine the spectral features of the pixels with the Gabor spatial features, and then input them into the fully connected layer. The output of the fully connected layer is expressed as

$$F\left(T_{i,j}, T'_{i,j}\right) = f\left(W\left(T'_{i,j}\middle\|T_{i,j}\right) + b\right) \tag{9}$$

where

F is the output of the fully connected layer;

$T_{i,j}$, $T'_{i,j}$ are the spectral and spatial characteristics;

W, b are the weight and bias of the fully connected layer.

Different directions emerge for dealing with this imbalanced problem, such as preprocessing of imbalanced data [46] and modification of classifiers [47]. Typically, sampling methods can alter imbalanced datasets into balance distributions. Studies have exhibited that a balanced distribution can achieve

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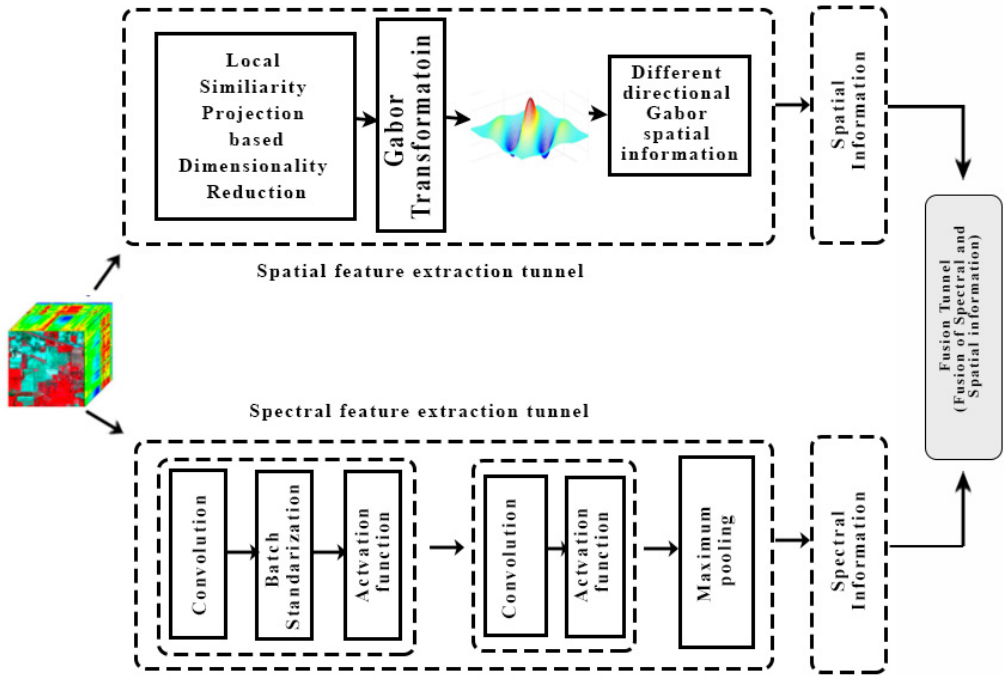


Fig. 2. Spatial–spectral dimension reduction.

better classification performance than an imbalanced distribution [48]–[50]. For large-size sets, methods, such as NearMiss1, NearMiss2, and NearMiss3, are proposed in [51], which are KNN undersampling algorithms for selecting some representative samples from majority sample sets. Due to the fact that features extracted from existing training samples are limited, repetitive or redundant training samples have great occurrence possibility in large-size sets. For small-size sets, algorithms, such as synthetic minority oversampling technique (SMOTE) [52], are developed. For each minority sample, the SMOTE randomly selects one of the KNN, then multiplies the corresponding difference vector with a random value chosen from 0 to 1, and finally adds this vector to the sample to generate an artificial one. Let x_1 and x_2 show the samples of sample classes, then $x_s = x_1 + (x_2 - x_1) \times \delta$ is an artificial sample created via local sampling method, where $\delta \in [0, 1]$ is a random number. Obviously, the artificial samples are linear combinations of actual ones that may not convey new features in the existing samples.

Finally, the joint spatial–spectral feature (F_{hsi}) is input into the two-layer classification [31], and the predicted probability distribution is

$$P(i,j) = \frac{1}{\sum_{j=1}^{Q} \exp(\theta_j T(F_{\text{hsi}}))} \begin{bmatrix} \exp(\theta_1 T(F_{\text{hsi}})) \\ \exp(\theta_2 T(F_{\text{hsi}})) \\ \vdots \\ \exp(\theta_c T(F_{\text{hsi}})) \end{bmatrix} \tag{10}$$

where

θ_j is the weight of the prediction layer in the jth column;

Q is the number of classes;

$P(i, j)$ is the 1-D vector that predicts the test pixel label $P(i, j) \in R^c$.

In the Hilbert space, the feature vector $x_i = (x_i^s, x_i^w)$ of the sample point is defined, which is composed of the spectral domain feature vector $x_i^w \in R^{N_w}$ of the sample point and the spatial characteristic vector $x_i^s \in R^{N_s}$ [32].

D. DCNN and Dual Optimization Classifier

On the DCNN, the use of spatial–spectral joint information to process HSIs can weaken the influence of different spectra or external objects of the same spectrum to some extent. The DCNN and dual optimization classifier models are shown in Fig. 3.

In the C1 layer, a batch size of 5×5 convolution kernel is used to convolve the input image, and the output size is 23 pixels $\times$ 23 pixels (the size of the training sample input in this article is 27 pixels $\times$ 27 pixels) of 50 feature maps. In the S1 layer, a 2×2 sampling window is used to downsample the maximum pooling of C1, and the output size is 11 pixels $\times$ 11 pixels of 50 feature maps. In the C2 layer, a 5×5 convolution kernel is used to convolve the input image S1, and the output size is 50 feature maps of 7 pixels $\times$ 7 pixels. In the S2 layer, a 2×2 sampling window is used to downsample the maximum pooling of C2, and 50 feature maps with a size of 3×3 pixels are output.

DCNN includes a convolutional layer and a downsampling layer. In the CNN, the convolutional layer is a feature extraction layer, through the convolution operation, which can enhance the original signal characteristics and reduce noise. Convolution operation can sharpen and blur the edge detection of the image [33].

Its convolution process is expressed as

$$a_{j,l} = f\left(\sum_{i \in M_j} a_{j,l-1} * k_{i,j,l} b_{j,l}\right) \tag{11}$$

where

$a_{j,l}$ is the activation value of output feature map j in layer l;

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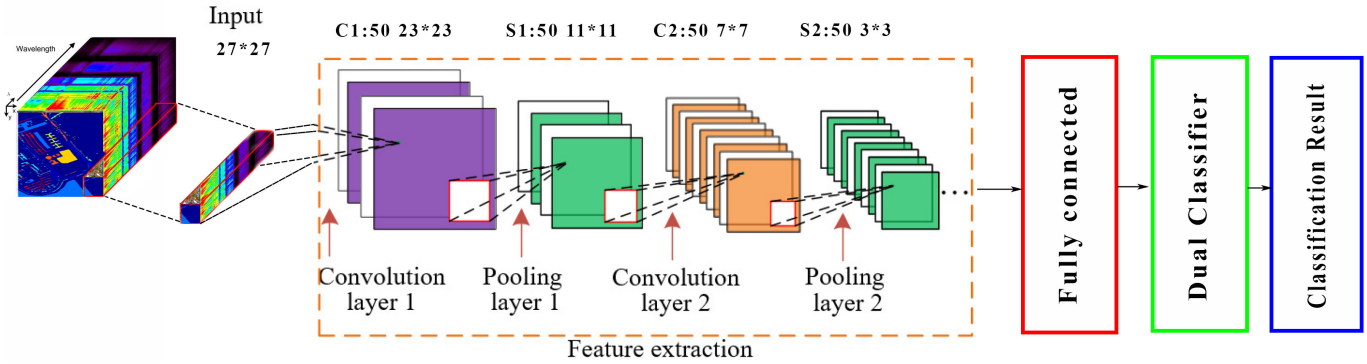


Fig. 3. DCNN and dual classifier.

$k_{i,j,l}$ is the input feature map i in layer $l-1$ and layer l;
The output feature map in j connected kernel
$b_{j,l}$ is the additive deviation from the output feature map j in layer l;
$f(\cdot)$ is the ReLu function;
M_j is the characteristic map j of linear map M.

The downsampling layer is also called the subsampling layer or the pooling layer, which can be regarded as a fuzzy filter. According to the principle of image local correlation, the image features are subsampled to achieve the effect of feature extraction again, which reduces the amount of data processing. It also retains the most valuable information. The forward propagation of the downsampling layer is expressed as

$$a_{j,l} = f(\beta_{j,l}\text{down}(a_{j,l-1}) + b_{j,l}) \tag{12}$$

where
down$(\cdot)$ is the subsampling function;
$\beta_{j,l}$ is the multiplicative bias of the output feature map j in the l layer.

Usually, this function adds up each different patch of the input feature map so that the output feature map is n times in both spatial dimensions. Each output graph gives its multiplication deviation and addition deviation [35].

In hyperspectral data, there is often a serious imbalance between the numbers of classes. If weight constraints are not imposed on various types of samples, then the trained network is more beneficial to samples with large amounts of data, but it is not good for samples with small amounts of data, as it results in insufficient learning of small samples and extremely low classification accuracy of small samples. To solve this problem, we adopted weight constraints on various samples. Therefore, different weights are assigned to various types of samples to impose constraints. To strengthen the learning ability of small samples, a larger weight is added before the small sample category and a smaller weight is added before the large sample category [31], then

$$L(W,b) = \frac{1}{2mn}\sum_{i=1}^{c}\mu_i\sum_{j=1}^{c_i}\left\|y_j^{c_i} - \hat{y}_j^{c_i}\right\|^2 + \frac{\lambda}{2}W^T W \tag{13}$$

where
μ_i is used to constrain the weights of different classes;
L is the deformation of L_2 norm;
c_i selects the number of sample points;
λ, m, n are constants;
W is the weight matrix;
$y_j^{c_i}$ selects the characteristic value of the sample;
$\hat{y}_j^{c_i}$ j is the approximate feature selection of samples.

It should be noted that DCNN, like other DL models, can continuously improve the classification accuracy of the model as the number of iterations increases. The dual classifier uses a two-layer stack; the first layer of reconstructed features is used as the second layer of the training set; and the second layer uses the output of the previous level as the training set. The dual classifier can fuse the original feature and the new feature, which is equivalent to feature expansion, and then standardize and normalize. This is the feature quantity that is composed of the feature dimension of the single-layer classifier and the output posterior probability. The classification is not the same, using a dual classifier will improve the classification accuracy.

III. Experimental Results and Discussion

In this section, we first define the dataset of HSI with its features, then we describe the parameter settings, then show the evaluation metrics, and finally different experimental results are carried out to test our proposed algorithm's classification accuracy and evaluate its classification performance by comparing it with some state-of-the-art classifiers.

A. Dataset

To ensure the effectiveness of the algorithm proposed in this article, the three standard datasets adopted in this experiment are all recognized in the field of HSI classification. They are the Indian Pines, Pavia University, and Salinas datasets. The three include many types of features and are universal, so they are widely used by scholars in testing. Details about the total number of samples within each class are mentioned in Table I.

1) Indian Pines: The Indian Pines dataset is the HSI data collected by the AVIRIS hyperspectral sensor on the Indiana farm in the United States. In 1992, the airborne visible infrared imaging spectrometer was used to image the Indian pine forest in Indiana, and then the size of 145 × 145 was intercepted and marked as use for the HSI classification test [38]. The AVIRIS spectrometer continuously images ground objects in 220 continuous bands, but the 104th to 108th, 220th, and 150th to 163th cannot be reflected by water. Therefore, it is

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TABLE I
NO. OF SAMPLES IN EACH DATABASE WITHIN EACH CLASS OF INDIAN PINES, PAVIA UNIVERSITY, AND SALINAS

Indian Pines			Pavia University			Salinas		
Class	Feature Type	Each class total Samples	Class	Feature Type	Each class total Samples	Class	Feature Type	Each class total Samples
C-1	Stone / Steel Tower	95	C-1	Broccli Green Weed 1	2009	C-1	Asphalt Road	6631
C-2	Hay / stock pile	489	C-2	Broccli Green Weed 2	3726	C-2	Meadow	18649
C-3	Corn corn seedlings	834	C-3	Falloli Fallow Land	1976	C-3	Crushed Stone	2099
C-4	Uncultivated C4 Uncultivated soybean land	968	C-4	Pear li Fallow Land	1394	C-4	trees	3064
C-5	Alfalfa C5 Alfalfa	54	C-5	Levelli Fallow Land	2678	C-5	metal plate	1345
C-6	Whole C6 Whole soybean field	614	C-6	Stubbli C Stubble	3959	C-6	bare soil	5029
C-7	Grassland Pasture	497	C-7	Celerli C Celery	3579	C-7	bitumen	1330
C-8	Firewood C8 Firewood	1294	C-8	Wild li Wild Grape	11271	C-8	Brick	3682
C-9	Construction/grass machine	380	C-9	Vineyli Cultivation Soil	6203	C-9	shadow	947
C-10	Pasture / pasture built	26	C-10	Old Cli Green Weed	3278			
C-11	Corn C11 Corn	234	C-11	Fourweek Roman Lettuce 1068	1068			
C-12	oats C12 oats	20	C-12	May Wli Romaine Lettuce	1927			
C-13	Uncultivated C13 Uncultivated corn field	1434	C-13	6-Weeli Romaine Lettuce	916			
C-14	soybean soybean seedling	2468	C-14	Sevenli Romaine Lettuce	1070			
C-15	Grass Tree	747	C-15	WildVineyard 7268	7268			
C-16	Wheat C16 Wheat	212	C-16	Grapeli Grape Rack	1807			

generally eliminated in the experiment. In the experiment, the 200 bands remaining after the elimination are studied [39]. AVIRIS is about a 20-m image imaging spatial resolution. Because mixed pixels are easy to produce, the classification difficulty is also relatively large.

2) Pavia University Data: The second HSI dataset is PaviaU, which was taken at the University of Pavia in Italy in 2002 by the ROSIS-3 sensor. Its spatial size is 610 × 340, the spatial resolution is 1.3 m, and the wavelength range is 0.4–0.86 μm. After removing 12 noise bands, 103 bands are finally left. Fig. 4 shows the pseudocolor composite map of the data from the University of Pavia and the real surface feature distribution map. The image data provide 42 776 labeled samples and nine types of features including trees, asphalt roads, and bricks, including plots, pastures, and other ground features.

3) Salinas Dataset: This dataset was also taken with the AVIRIS imaging spectrometer, similar to the Indian Pines image, which mainly imaged the Salinas Valley in the United States. But it is different from Indian Pines in that the Salinas dataset can reach a spatial resolution of 3.7 m. The original image contains 224 bands, and the 108–112th, 154–167th, and 224th bands that cannot be reflected by water need to be eliminated. The remaining band images are 204. With an image size of 512 × 217 and a total of 56 975 background pixels, only 54 129 pixels can be used in the classification experiment, and the pixels can be divided into 16 categories [42].

B. Evaluation Metrics

After the HSI is classified, we need to evaluate the results of its classification. Commonly used classification results evaluation indicators are overall accuracy (OA), average accuracy (AA), and kappa coefficient. These indicators are usually calculated based on the confusion matrix. Assuming that the HSI contains C-type features, the confusion matrix A is a $C \times C$ matrix, where A_{ij} represents the number of test samples of type i divided into type j, then the total number of test samples $N = \sum_i^C \sum_j^C A_{ij}$, and then

$$\mathrm{OA} = \frac{\sum_i^C A_{ii}}{N} \times 100\% \tag{14}$$

$$\mathrm{AA} = \frac{\sum_i^C CA_i}{C} \times 100\% \tag{15}$$

$$k = \frac{N \times \sum_i^C A_{ii} - \sum_i^C \left(\sum_i^C A_{ij} \times \sum_j^C A_{ji}\right)}{N^2 - \sum_i^C \left(\sum_j^C A_{ij} \times \sum_j^C A_{ji}\right)}. \tag{16}$$

The OA, AA, and kappa coefficient are used as the classification performance evaluation indicators. Among them, OA represents the percentage of test samples that are correctly classified, AA is the average accuracy of each class on the test samples, and the kappa coefficient is an index to measure the accuracy of classification. In the experiment, 20% pixels in each category of the dataset are randomly selected as the training sample, and the remaining pixels are used as the

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TABLE II
PARAMETERS' SETTING USED DURING THE DESIGN OF THE LSPGF FOR EACH DATABASE AND CONFIGURATIONAL LAYERS OF NEURAL NETWORKS

	Indian Pines	Pavia University	Salinas
Layer c1 n^{c1} X k^{c1} X k^{c1}	5*5 (Output 23 × 23 pixels)	5*5 (Output 23 × 23 pixels)	5*5 (Output 23 × 23 pixels)
Layer s1 Pooling	2*2 (Output 11 × 11 pixels)	2*2 (Output 11 × 11 pixels)	2*2 (Output 11 × 11 pixels)
Layer c2 n^{c2} X k^{c2} X k^{c2}	5*5 (Output 7 × 7 pixels)	5*5 (Output 7 × 7 pixels)	5*5 (Output 7 × 7 pixels)
Layer s2 Pooling	2*2 (Output 3 × 3 pixels)	2*2 (Output 3 × 3 pixels)	2*2 (Output 3 × 3 pixels)
Layer *fc*1 k^{fc1}	c	c	c
Activation	*ReLU*	*ReLU*	*ReLU*
Layer *fc*2 k^{fc2}	c	c	c
Batch norm	Yes	Yes	Yes
Activation	Softmax	Softmax	Softmax

test sample. To avoid random deviation, each method repeats the experiment ten times and calculates the average result. Experiments were implemented with MATLAB 2018b and Python on a Windows 10 operation system and conducted on an Intel Core i7-8700 CPU 3.20-GHz desktop with a 16-GB memory (other settings are shown in Table II).

C. Comparison With Other Methods

To verify the effectiveness of the classification performance of our proposed algorithm, we compared it with other DLs having different classifiers or different methods of using spectral–spatial classification. We selected SVM [36], sparse representation-based classification (SRC) [37], SAE [38], CNN [11], 2-D-CNN [30], MFL [39], spectral–spatial residual network (SSRN) [40], and deep feature fusion network (DFFN) [41] for the purpose of comparison. Parameters setting for each methods is considered to be the same as that mentioned in their configuration setting.

The classification performances of all the selected methods for the three datasets are shown in Tables III–V and compared with our better-performing proposed algorithm. Classification results of maps are shown from Figs. 5 to 7, respectively.

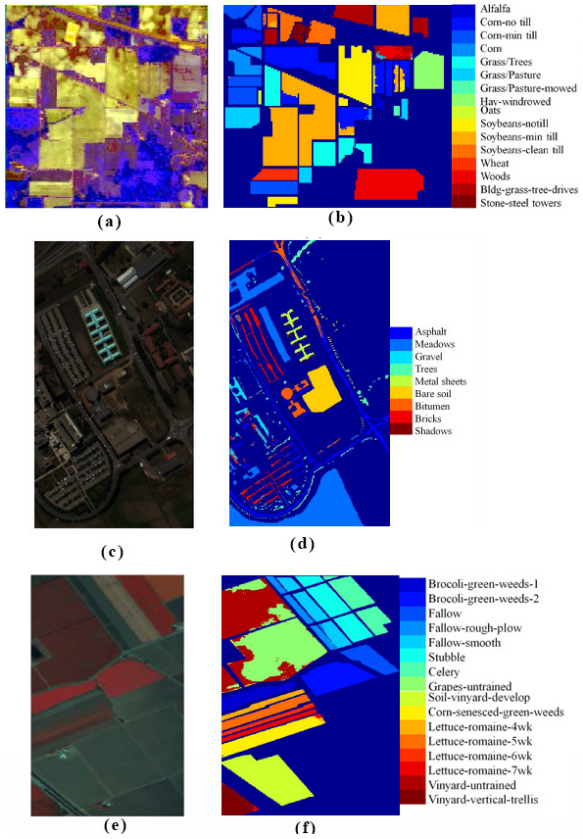


Fig. 4. (a) Indian pines false-color composite. (b) Indian pines ground truth with color codes. (c) PaviaU false-color composite. (d) PaviaU ground truth with color codes. (e) Salinas false-color composite. (f) Salinas ground truth with color codes.

1) Indian Pines Dataset: From Table III, it can be seen that the LSPGF algorithm performs well among all three evaluation metrics with 100% accuracy in two classes, and in all classes, the accuracy is almost more than 80%. Compared with the other algorithms, the proposed algorithm increased the AA from 66.80% to 93.88%, which is also 3% higher than 2-DCNN, and 6% higher than the other spectral–spatial classification algorithm SSRN. The results show that our algorithm is far better than other feature learning algorithms like MFL and classifiers such as SVM, SAE, and CNN.

From Fig. 5, it is visible that SRC, SVM, SAE, CNN, and MFL are not classifying as much accurately that shown in the ground-truth image, particularly in the center part of image, and the classes include corn-notill (SVM and SAE are poor), corn-mintill, soybean-notill, and soybean-mintill, while on the other side 2-DCNN, DFFN, and LSPGF have significantly uniform classifications of each class, especially in hay-windrowed, oats, and wheat. Compared to the DFFN, which is a dual feature fusion method for spatial–spectral features, LSPGF still has advantages in all classification metrics (OA, AA, and kappa), and still achieves 100% accuracy in some classes. This is because our algorithm increases the feature information more by getting local similarity among the pixels in the spatial domain during the data feature selection and incorporates more spatial information convolutional tunnel, not just texture features.

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TABLE III

INDIAN PINES' IMAGE DATASET CLASSIFICATION ACCURACY WITH DIFFERENT ALGORITHMS AND LSPGF ALGORITHM

CLASS	SVM	SRC	SAE	CNN	2D-CNN	MFL	SSRN	DFFN	LSPGF
1	83.30	27.27	36.53	7.95	78.05	91.63	82.93	**100.00**	87.66
2	41.70	53.54	68.04	70.69	79.77	91.87	92.37	80.00	**97.54**
3	37.40	47.78	59.75	52.84	83.94	95.36	**97.86**	93.44	95.46
4	70.00	44.88	32.37	27.51	90.14	80.14	62.44	**94.37**	89.88
5	54.30	78.21	77.16	90.44	**99.54**	91.73	97.47	91.00	90.16
6	80.70	83.53	94.45	98.59	98.63	98.14	99.85	99.24	**100.00**
7	93.40	75.00	32.17	41.15	76.00	**94.35**	60.00	68.00	93.87
8	85.90	89.66	98.62	99.96	99.77	99.00	**100.00**	99.77	96.89
9	80.00	42.86	22.22	16.32	100.00	85.00	44.44	88.89	**91.55**
10	48.00	60.53	62.46	67.84	89.24	89.10	91.30	**93.82**	85.16
11	65.30	68.67	81.51	78.16	95.74	95.89	**97.69**	94.15	90.15
12	32.40	48.31	47.98	42.01	76.55	85.99	**91.37**	84.80	89.57
13	96.40	85.15	91.23	98.97	**100.00**	99.59	**100.00**	97.83	97.15
14	66.50	90.09	88.82	97.65	99.21	99.17	98.59	97.25	**100.00**
15	35.90	48.55	57.88	62.62	85.30	88.11	82.71	96.54	**97.55**
16	97.60	95.06	86.21	76.02	**100.00**	88.39	**100.00**	98.80	99.47
OA (%)	68.70	67.42	74.38	75.47	91.31	93.91	94.77	95.01	**98.91**
AA (%)	66.80	64.94	64.84	64.30	90.74	92.09	87.44	92.37	**93.88**
KC	63.12	62.93	70.31	71.80	90.06	93.04	94.03	94.31	**95.16**

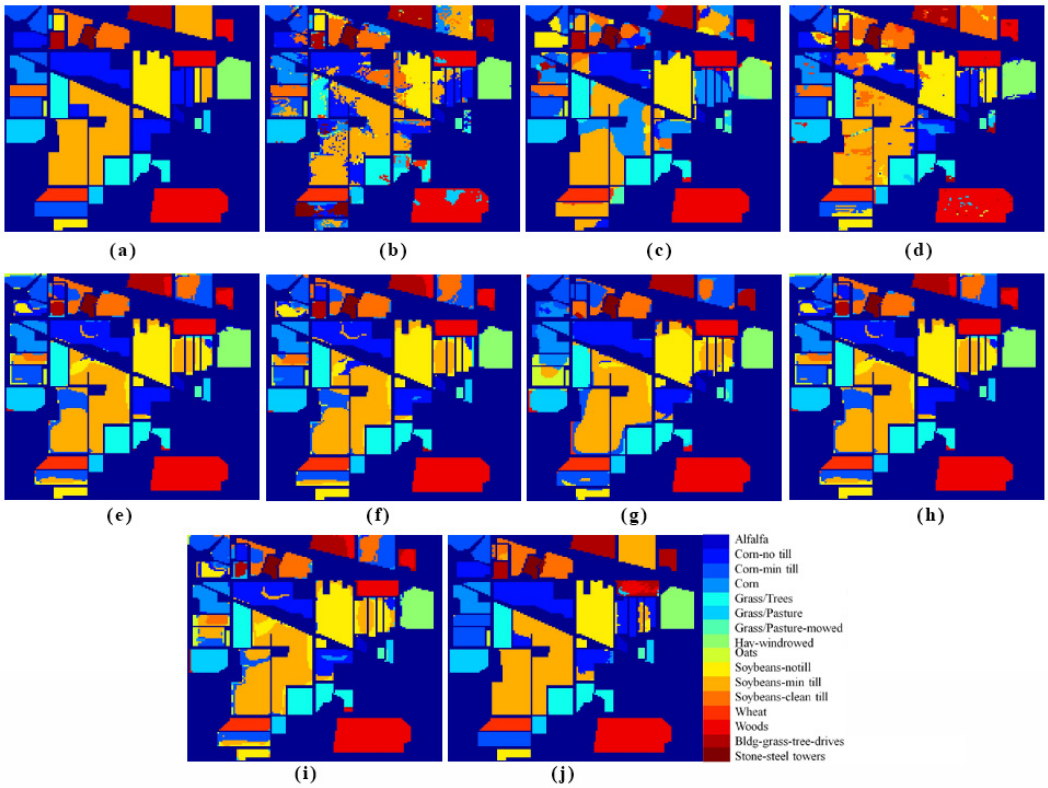


Fig. 5. Indian Pines image. (a) Ground truth. (b) SVM (OA = 68.70%). (c) SRC (OA = 67.42%). (d) SAE (OA = 74.38%). (e) CNN (OA = 75.47%). (f) 2-D-CNN (OA = 91.31%). (g) MFL (OA = 93.91%). (h) SSRN (OA = 94.77%). (i) DFFN (OA = 95.01%). (j) Proposed (OA = 98.91%).

2) Salinas Dataset: The statistical classification results of the Salinas dataset are shown in Table IV. As can be seen in Table IV, SAE, 2-D-CNN, SSRN, and the proposed algorithm have better results than SVM, SAE, and CNN, and the reason is that these methods extract spatial information through local connections and reduce network parameters

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TABLE IV
SALINAS' IMAGE DATASET CLASSIFICATION ACCURACY WITH DIFFERENT ALGORITHMS AND LSPGF

CLASS	SVM	SRC	SAE	CNN	2D-CNN	MFL	SSRN	DFFN	LSPGF
1	98.61	98.69	97.09	97.34	100.00	99.84	**100.00**	99.49	97.55
2	99.07	98.72	99.54	99.29	98.63	99.49	87.16	**99.81**	99.13
3	87.52	96.14	96.65	96.51	94.37	99.96	99.48	96.07	**100.00**
4	99.20	99.17	98.15	**99.66**	98.17	98.24	99.63	89.99	96.45
5	88.15	94.03	99.05	96.97	77.06	98.84	**99.39**	99.35	**100.00**
6	99.53	99.74	99.96	99.60	**100.00**	99.71	**100.00**	99.90	98.16
7	99.47	99.41	98.89	99.49	99.91	99.83	**100.00**	99.63	96.13
8	86.71	72.83	85.14	72.25	87.93	90.44	88.26	**91.44**	92.15
9	98.98	98.16	99.47	97.53	**100.00**	99.86	**94.15**	94.16	97.32
10	93.08	89.64	91.60	91.29	95.83	**98.20**	94.15	95.98	92.16
11	95.21	94.75	92.74	97.58	99.33	91.72	**99.90**	98.37	95.54
12	99.63	99.49	99.30	**100.00**	98.94	99.81	97.15	**100.00**	99.71
13	97.36	97.25	97.90	99.02	99.89	99.21	**100.00**	98.55	99.10
14	93.99	92.94	93.56	95.05	97.52	87.42	99.33	**99.52**	96.15
15	68.45	62.52	70.60	76.83	91.01	71.32	93.30	88.46	**97.16**
16	97.43	96.94	92.69	98.94	89.21	98.12	99.15	**99.32**	97.15
OA (%)	89.92	93.15	94.52	94.84	95.49	95.75	96.35	96.76	**97.25**
AA (%)	93.90	93.15	94.52	94.83	95.49	95.75	96.94	96.88	**97.12**
KC (%)	91.15	92.93	90.31	81.80	90.06	93.04	94.03	94.31	**95.16**

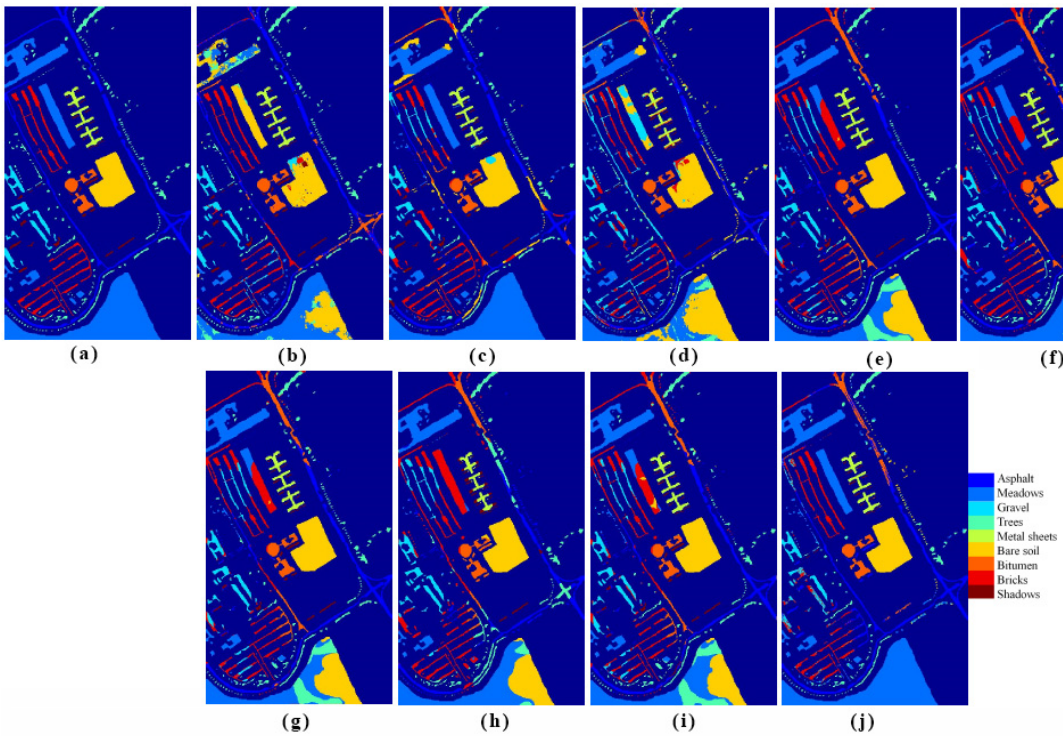


Fig. 6. University of Pavia dataset. (a) Ground truth. (b) SVM (OA = 94.92%). (c) SRC (OA = 93.15%). (d) SAE (OA = 94.52%). (e) CNN (OA = 94.84%). (f) 2-D-CNN (OA = 95.49%). (g) MFL (OA = 95.75%). (h) SSRN (OA = 91.54%). (i) DFFN (OA = 96.76%). (j) Proposed (OA = 97.25%).

through weight-sharing. In many classes, the results of SVM are not very accurate in classifications and SRC does not perform well in that dataset. SSRN has much better classification accuracy in many classes because of more spectral–spatial information in the bands. Our proposed algorithm is still performing well with an increase of 5% in AA and a 3% increase

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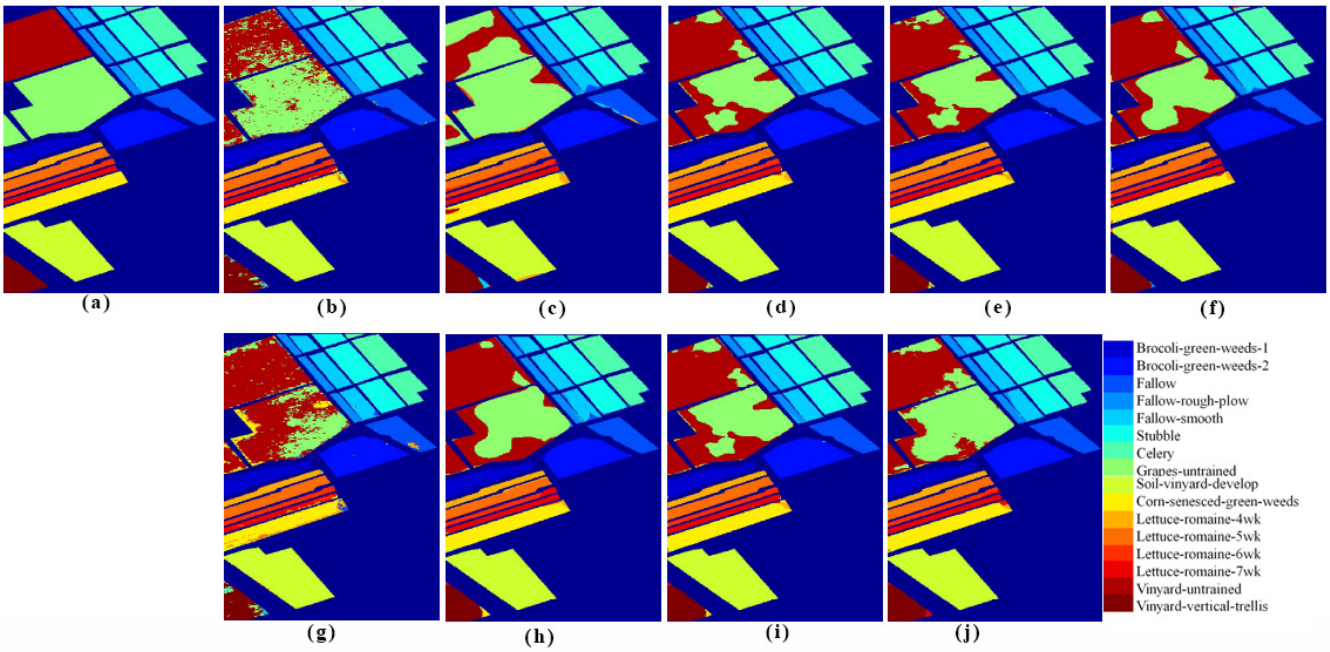


Fig. 7. Salinas dataset. (a) Ground truth. (b) SVM (OA = 89.92%). (c) SRC (OA = 93.15%). (d) SAE (OA = 94.52%). (e) CNN (OA = 94.84%). (f) 2-D-CNN (OA = 95.49%). (g) MFL (OA = 95.75%). (h) SSRN (OA = 96.35%). (i) DFFN (OA = 96.76%). (j) LSPGF (OA = 97.25%).

from CNN and a 2% increase from 2-D-CNN and MFL. The results of SSRN and DFFN are 1% lesser than the proposed algorithm because of their similar fusion approach to spectral and spatial information extraction. Fig. 7 shows that SVM method classification is not much better in visual maps, while SSRN, DFFN, and the proposed algorithm are much better in removing the noise from most of the classes of a dataset.

3) PaviaU Dataset: Similar conclusions are drawn from the PaviaU dataset as shown in Table V. Overall, the increase in AA is around 8% and the proposed algorithm performs better than DFFN and SSRN by 2%. Classification maps are shown in Fig. 8, which clarify that the class-wise visual representation of the proposed algorithm is much better and visible. Class-wise accuracy is more than 90% in every class which provides effective results and proves that the proposed algorithm is much better in solving the spectral–spatial problem together.

The algorithm in this article achieves a better classification of the two datasets. The overall classification accuracy of the Indian Pines dataset is 98.91% and the overall classification accuracy of the University of Pavia dataset is 97.25%. Compared with other methods, this LSPGF algorithm has an average classification accuracy and kappa coefficient than the three datasets and has been greatly improved, which fully verifies the effectiveness of the proposed algorithm for HSI classification.

It can be seen from Figs. 5 to 7 that the algorithm in this article has a good classification effect on hyperspectral data with many vegetation, complex distribution, and unbalanced datasets, and can effectively remove the salt and pepper phenomenon [25], which verifies that the algorithm in this article has a good effect on vegetation in HSIs.

4) Better Classification Performance: The spectral curves of asphalt roads and asphalt, metal plate surfaces, and shadows in the University of Pavia dataset are very similar. As shown in Fig. 5, the classification effect of this algorithm is better than other algorithms, which verifies that this algorithm is better for ground objects with similar spectral characteristics.

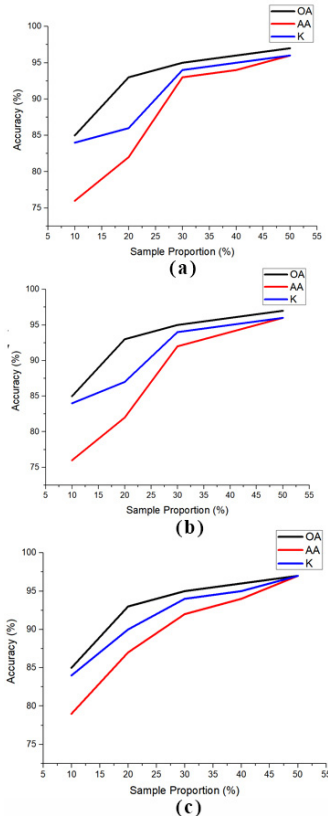


Fig. 8. Different no. of training samples and accuracy comparisons. (a) Indian Pines. (b) Pavia University. (c) Salinas.

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TABLE V
UNIVERSITY OF PAVIA'S IMAGE CLASSIFICATION ACCURACY WITH DIFFERENT ALGORITHMS AND LSPGF

CLASS	SVM	SRC	SAE	CNN	2D-CNN	MFL	SSRN	DFFN	LSPGF
1	80.89	63.07	87.73	**99.68**	93.63	81.32	91.16	94.24	93.11
2	81.07	79.66	94.91	**99.87**	98.2	90.91	90.16	99.1	97.32
3	77.6	68.14	93	96.31	92.32	85.09	95.77	86.49	**97.15**
4	96.36	92.05	**98.12**	81.99	99	96.61	95.07	97.57	93.15
5	99.48	99.75	**100**	**100**	89.99	99.63	**100**	98.33	90.15
6	80.53	88.15	90.66	80.16	**99.39**	94.33	98.94	81.16	91.16
7	85.43	86.73	92.61	95.63	94.93	95.94	93.32	87.87	**96.11**
8	75.06	72.47	91.47	**99.35**	83.76	82.65	87.16	92.82	90.01
9	98.38	97.99	90.85	88.16	91.65	99.79	88.1	98.92	**100**
OA (%)	94.92	93.15	94.52	94.84	95.49	95.75	91.54	96.76	**97.25**
AA (%)	86.09	83.11	93.26	93.46	93.65	91.81	93.30	92.94	**94.24**
KC (%)	87.15	78.93	89.31	91.80	90.06	93.04	94.03	94.31	**95.16**

TABLE VI
TOTAL CONSUMPTION OF TRAINING AND TESTING TIME (IN S) OF ALL METHODS IN DIFFERENT DATABASES (12% SAMPLES OF EACH CLASS)

Time (seconds)	Indian Pines		Pavia		Salinas	
Methods	Training time	Testing time	Training time	Testing time	Training time	Testing time
SVM	31.01	2.71	35.97	3.14	32.73	2.86
SRC	39.12	3.65	45.38	4.23	41.30	3.85
SAE	52.14	4.13	60.48	4.79	55.04	4.36
CNN	66.14	3.15	76.72	3.65	69.82	3.33
2D-CNN	87.87	3.19	101.93	3.70	92.76	3.37
ELM	89.12	5.15	103.38	5.97	94.08	5.44
SSRN	56.14	6.71	65.12	7.78	59.26	7.08
DFFN	87.02	3.99	100.94	4.63	91.86	4.21
Proposed	101.14	5.61	117.32	5.51	106.76	5.92

The algorithm in this article uses 2-D Gabor spatial feature extraction, which can ensure that low-level features such as shape, texture, and edges do not change much, and use the DCNN framework for hyperspectral vegetation classification, which can obtain comparisons in limited training tags. High classification accuracy can effectively remove the salt and pepper phenomenon. HSI features have rich spectral characteristics and similar spectra. The experimental results show that the algorithm in this article can effectively classify features with similar spectral characteristics by using spatial–spectral fusion information, which makes up for the use of spectral information or spatial information efficiently together.

D. Impact of Performance Parameters

1) Execution Time: Executing time of different algorithm comparisons is shown in Table VI, with different datasets for random 12% samples in each class. It can be seen that our proposed algorithm uses multiple steps like local similarity-based dimension reduction, Gabor filtering in 2-D, and CNN with fully connected layers, yet our algorithm is better than SSRN. SVM is used as a pixel-based classification and that is why the reason time of classification is high; CNN and 2-DCNN have fewer layers than our proposed methods so the execution time is higher than the proposed algorithm.

TABLE VII
SPATIAL SIZE IMPACT ON OA WITH DIFFERENT DATABASES

OA	3 X 3	5 X 5	7 X 7	9 X 9	11 X 11
Indian Pines	86.81	93.15	95.91	97.16	96.14
PaviaU	89.14	94.71	95.91	96.13	95.41
Salinas	87.61	95.13	96.15	97.19	96.01

2) Spatial Size Effect: It is important to find the best HSI cube to be explored, and performance needs to be evaluated on each dataset. We tested the spatial–spectral features with different sizes: 3×3, 5×5, 7×7, 9×9, and 11×11 for each of the 12% samples in each class and computed OA (as shown in Table VII).

The spectral–spatial features that were extracted helped decrease the overall interference among the variance of classes, and caused an increase in the classification results. However, an oversized block may cause an increase in additional oversize noise in certain blocks, especially in corners or places where the class category is changed. Therefore, in our experiments, a spatial block size of 5×5 is used to avoid noise issues in results.

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TABLE VIII
SELECTIVE PARAMETERS' IMPACT ON OA WITH DIFFERENT HSI DATABASES

OA	LSP + CNN	LSP + GF	LSPGF
Indian Pines	95.26	95.16	96.39
PaviaU	93.54	93.12	94.18
Salinas	94.01	93.34	94.81

3) No. of Training Samples: In this section, with a spatial block size of 9×9 and with limited training samples, we evaluated the performance of our algorithm. Training samples were in different percentages of each class, from 10% to 50%, and OA was computed. It can be observed from Fig. 8 that after 40% of the samples the results achieved were almost similar.

The results in Fig. 8 show the output's impact of the training sample's relationship on the performance.

4) Different Selective Parameters: Further experiments were carried out by using selective parameters of the proposed model with 12% samples of the training set of each class, as shown in Table VIII. Partial features of the proposed algorithm are tested to evaluate the performance of the algorithm. Combinations are LSP + CNN in which GF is not used, and then LSP is applied with Gabor filtering and later compared with our proposed combined model.

The proposed algorithm performs better when selective features are reduced and the value of OA reduces with the reduced parameters. However, execution time will also reduce with the reduced parameters, which will impact the overall performance of the model.

IV. CONCLUSION

In this article, we have developed a better approach LSPGF for the spatial–spectral classification of remotely sensed HSIs. Use the LSP-based DR model to reduce the dimensionality of hyperspectral data through local protective discriminant analysis, eliminate the correlation between features, and reduce the feature dimension to obtain a clear spatial structure, and ensure that low-level features such as shape, texture, and edge have not changed much. The 2-D Gabor filter is used to extract spatial features; the spectral tunnel is used to extract spectral information; and the DCNN is used to reduce the number of parameters and improve the training performance of the algorithm by using the relative relationship of the spatial structure. Each convolutional layer in the CNN is followed by a secondary extraction downsampling layer. This unique secondary feature extraction structure reduces the feature resolution and also enables the network to have a higher input sample during feature extraction. The experimental results of the hyperspectral data show that the classification accuracy of hyperspectral remote sensing images is high, which verifies the good performance of the algorithm in this article in terms of classification accuracy.

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